Nagar YuwakShikshanSanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	**2026-27 (ODD)**	**Course Name**	**Lab : Embeded Systems**
Semester	**7**	**Course Code**	**23IOT1706**
Roll No	**61**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**08**
Course Outcome	**CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools.** **CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals.** **CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols.** **CO4: Design, simulate, and test embedded system applications for real-world applications.**
Aim	Write and execute an ARM7 assembly language program in Keil µVision to perform addition subtraction and multiplication of two numbers.
Theory (100 words)	ARM7 is a 32-bit RISC (Reduced Instruction Set Computer) processor architecture widely used in embedded systems and microcontroller applications. It follows a simple instruction set that allows arithmetic, logical, data transfer, and control operations to be performed efficiently. ARM7 contains multiple general-purpose registers, including R0 to R15, which are used to store data, addresses, and intermediate results during program execution. In this experiment, an ARM7 assembly language program is written and executed using **Keil µVision** to perform three basic arithmetic operations: **addition, subtraction, and multiplication** of two numbers. Initially, two numbers are loaded into registers R0 and R1 using the MOV instruction. For example, the values 20 and 10 can be stored in R0 and R1 respectively. The ADD instruction is used to add the contents of two registers. The instruction ADD R2, R0, R1 adds the values stored in R0 and R1 and stores the result in R2. Therefore, R2 contains 30. The SUB instruction is used to subtract one number from another. The instruction SUB R3, R0, R1 subtracts the value in R1 from R0 and stores the result in R3. For the given values, the result is 10. The MUL instruction performs multiplication of two register values. The

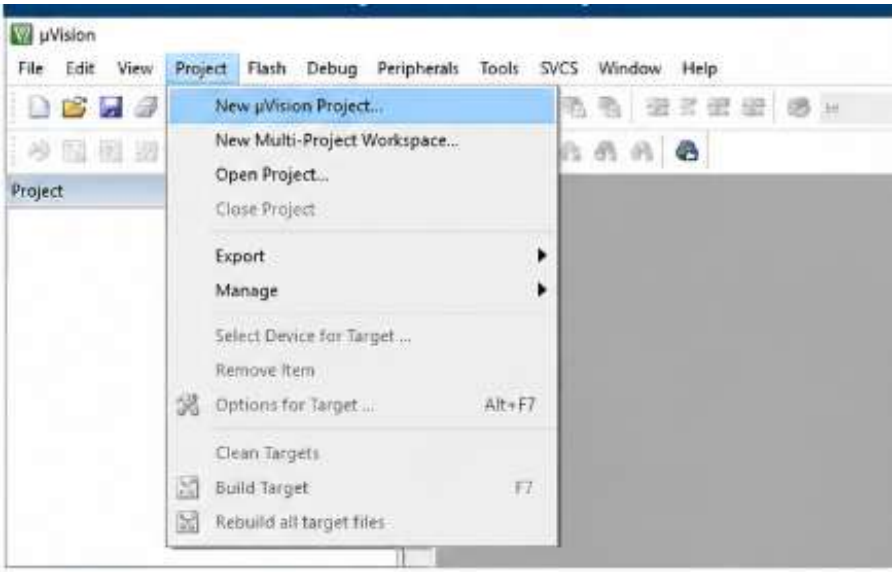
	instruction MUL R4, R0, R1 multiplies the values stored in R0 and R1 and stores the result in R4. Thus, for 20 and 10, the multiplication result is 200. The basic instructions used in the program are MOV, ADD, SUB, MUL, and B. The MOV instruction transfers an immediate value into a register. The ADD, SUB, and MUL instructions perform the required arithmetic operations. The B instruction is used to branch to a specified label and can create an infinite loop at the end of the program, preventing unwanted execution beyond the program. Keil µVision provides an integrated development environment for writing, assembling, building, and debugging ARM7 programs. After writing the assembly program, the project is built to check for syntax and compilation errors. The program can then be executed in debug mode using the step-by-step execution facility. The contents of the processor registers can be viewed through the register window. For this experiment, if the two input numbers are **20 and 10**, the expected results are **30 for addition, 10 for subtraction, and 200 for multiplication**. These results are stored in R2, R3, and R4 respectively. By observing these registers in the Keil debugger, the correct execution of the ARM7 assembly language program can be verified. Thus, this experiment demonstrates the basic arithmetic capabilities of the ARM7 processor and provides practical understanding of register-based operations, assembly language instructions, and program debugging in the Keil µVision environment.
Procedure and Execution (100rds)	1. Open **Keil µVision** and create a new project by selecting **Project → New µVision Project**. 2. Select an **ARM7-based microcontroller**, such as **LPC2148**, as the target device. 3. Create a new assembly source file with the .s extension and add it to **Source Group 1**. 4. Write the ARM7 assembly program to perform addition, subtraction, and multiplication of two numbers. 5. Load the two input numbers into registers **R0** and **R1** using the MOV instruction. 6. Use the ADD instruction to perform addition and store the result in **R2**. 7. Use the SUB instruction to perform subtraction and store the result in **R3**.

8. Use the MUL instruction to perform multiplication and store the result in **R4**.
9. Build the project using **Build Target (F7)** and check that there are no errors.
10. Start the debugging session using **Debug → Start/Stop Debug Session**.
11. Execute the program step-by-step using the **Step** option or **F11** key.
12. Open the **Registers** window and observe the values of R0, R1, R2, R3, and R4.
13. For input values **20 and 10**, the registers should show:

    R2 = 30 → Addition
    R3 = 10 → Subtraction
    R4 = 200 → Multiplication

14. Verify that the obtained register values match the expected arithmetic results and stop the debugging session after verification.

Stepwise Screenshots with steps:

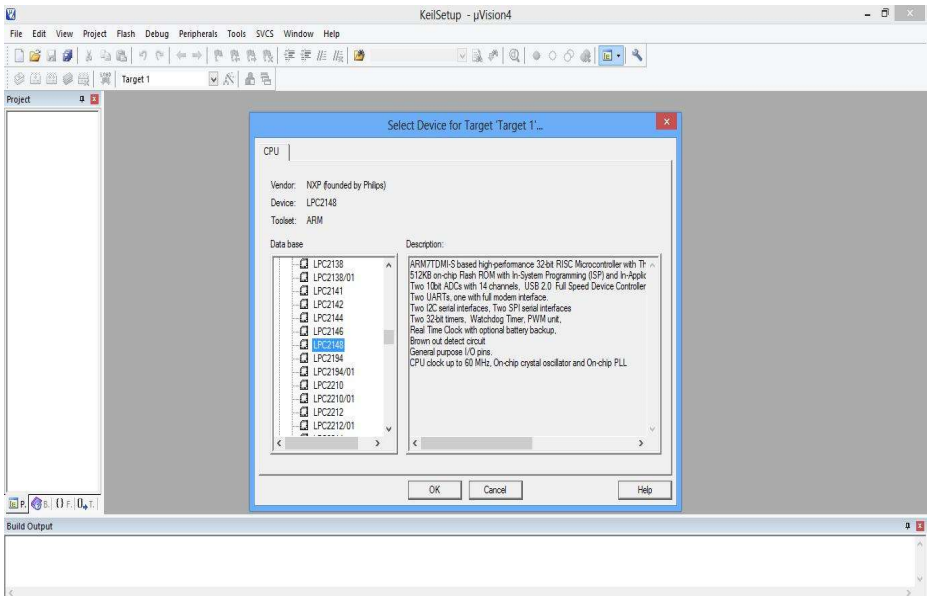


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```
; sheikhwasimuddin  cseiot61  es lab 8
        AREA    |.text|, CODE, READONLY
        ENTRY

START
        MOV     R0, #20             ; First number = 20
        MOV     R1, #10             ; Second number = 10

        ; Addition
        ADD     R2, R0, R1          ; R2 = 20 + 10  = 30

        ; Subtraction
        SUB     R3, R0, R1          ; R3 = 20 - 10  = 10

        ; Multiplication
        MUL     R4, R0, R1          ; R4 = 20 * 10  = 200

STOP
        B       STOP                ; Infinite loop
        END
```

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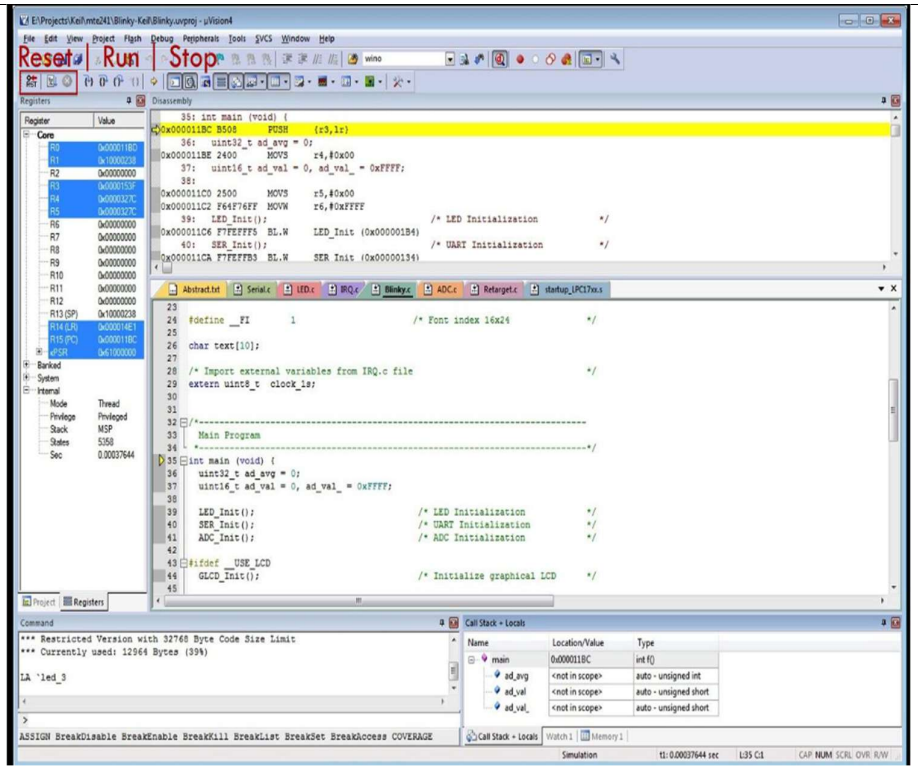
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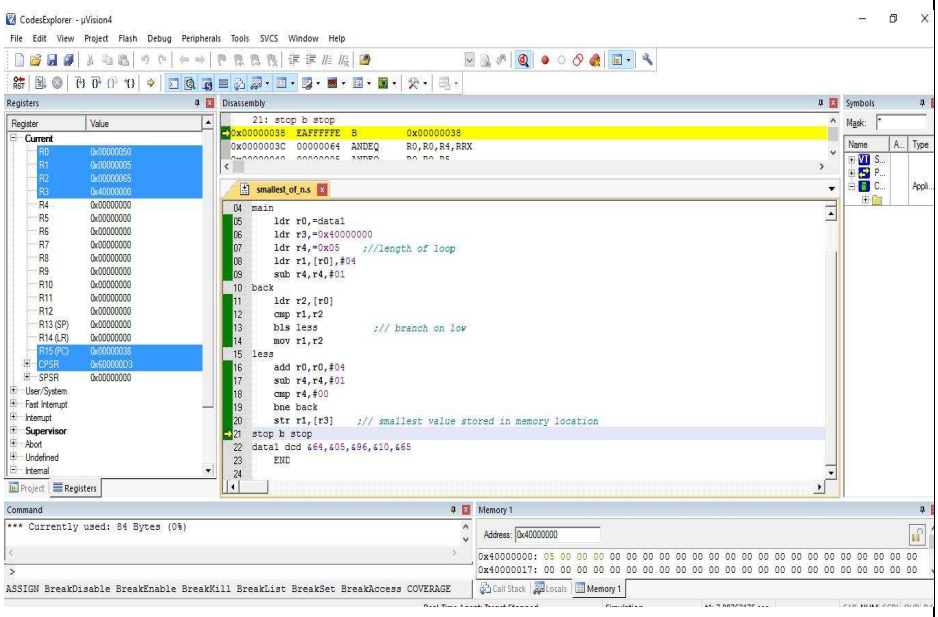
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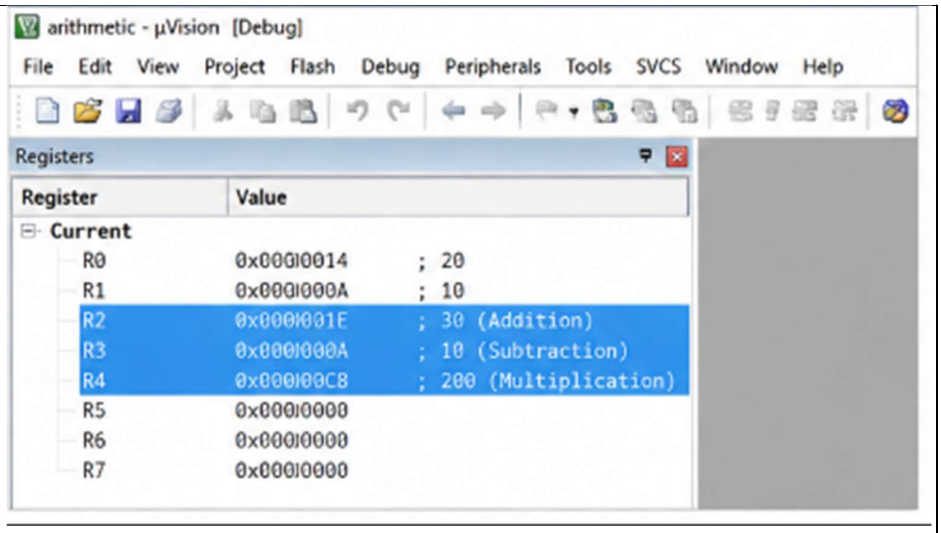
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<table>
<tr><td></td><td> 
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<tr><td>Conclusion</td><td>The ARM7 assembly language program was successfully written, assembled, and executed using the Keil µVision IDE. The program performed addition, subtraction, and multiplication of two numbers using the ADD, SUB, and MUL instructions. The results were stored in the respective ARM7 registers and verified using the Keil debugger. For the input values 20 and 10, the obtained results were 30 for addition, 10 for subtraction, and 200 for multiplication. Thus, the experiment successfully demonstrated basic arithmetic operations, register usage, and debugging in an ARM7-based system.</td></tr>
<tr><td>Date</td><td>10-09-2026</td></tr>
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